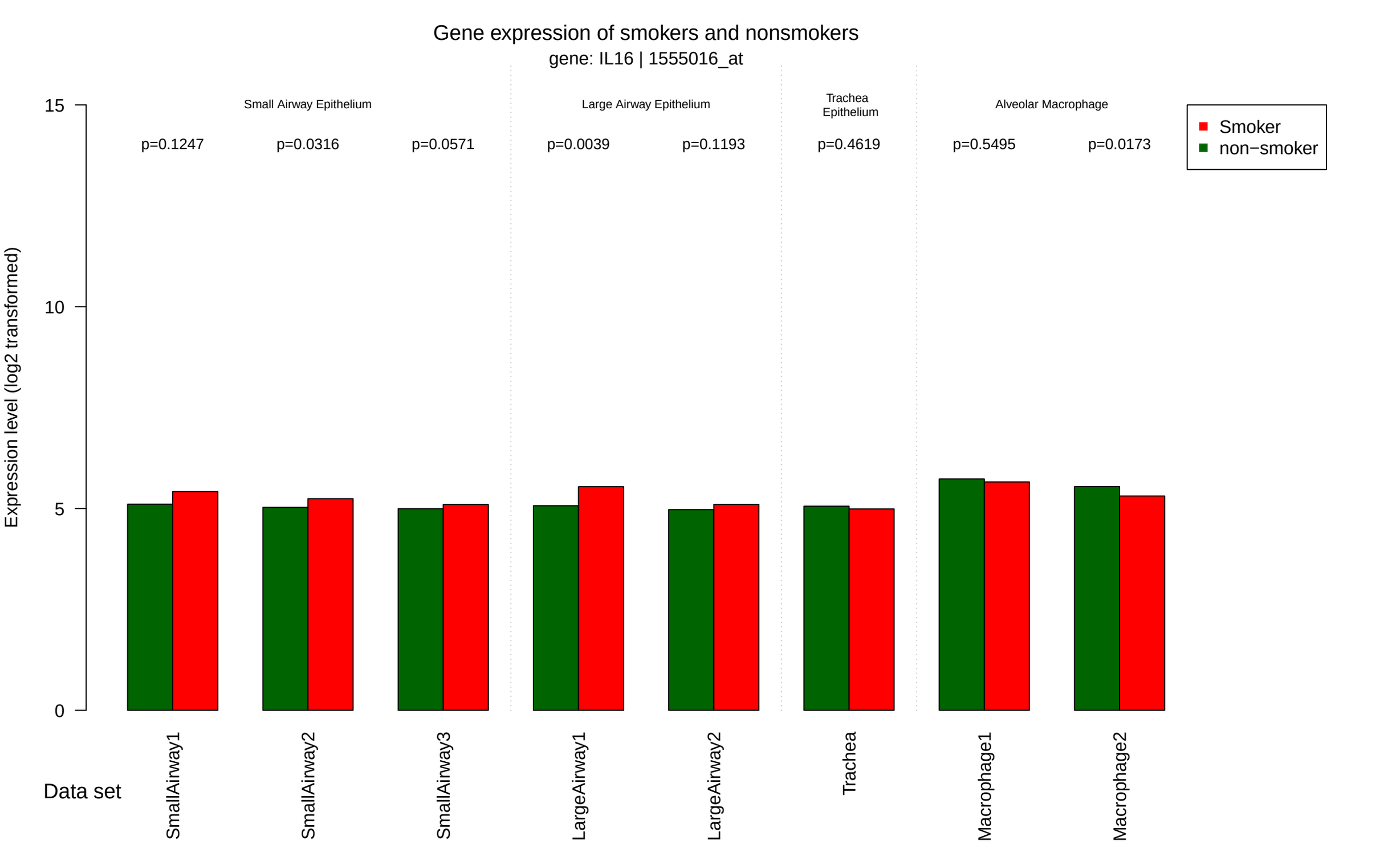
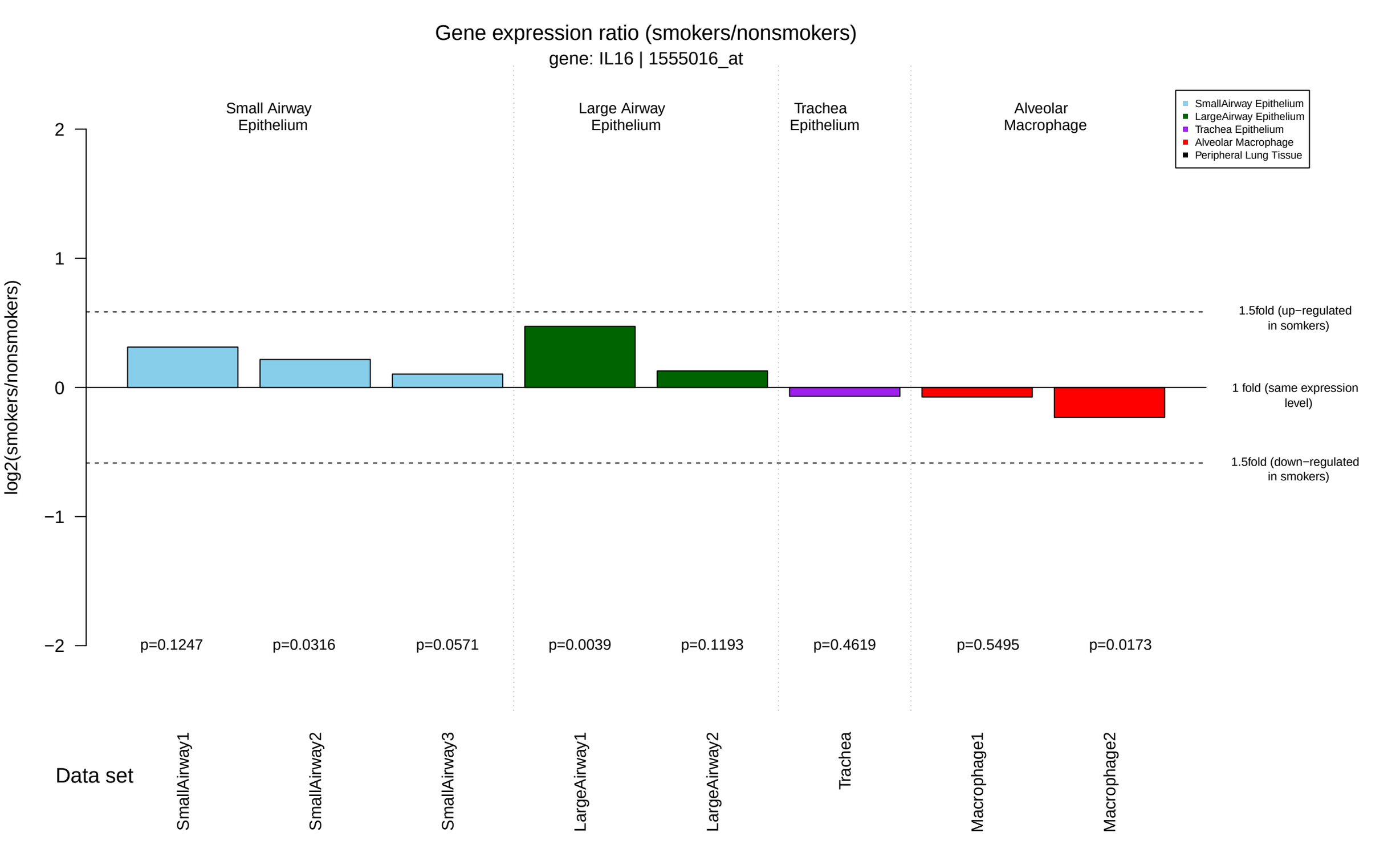
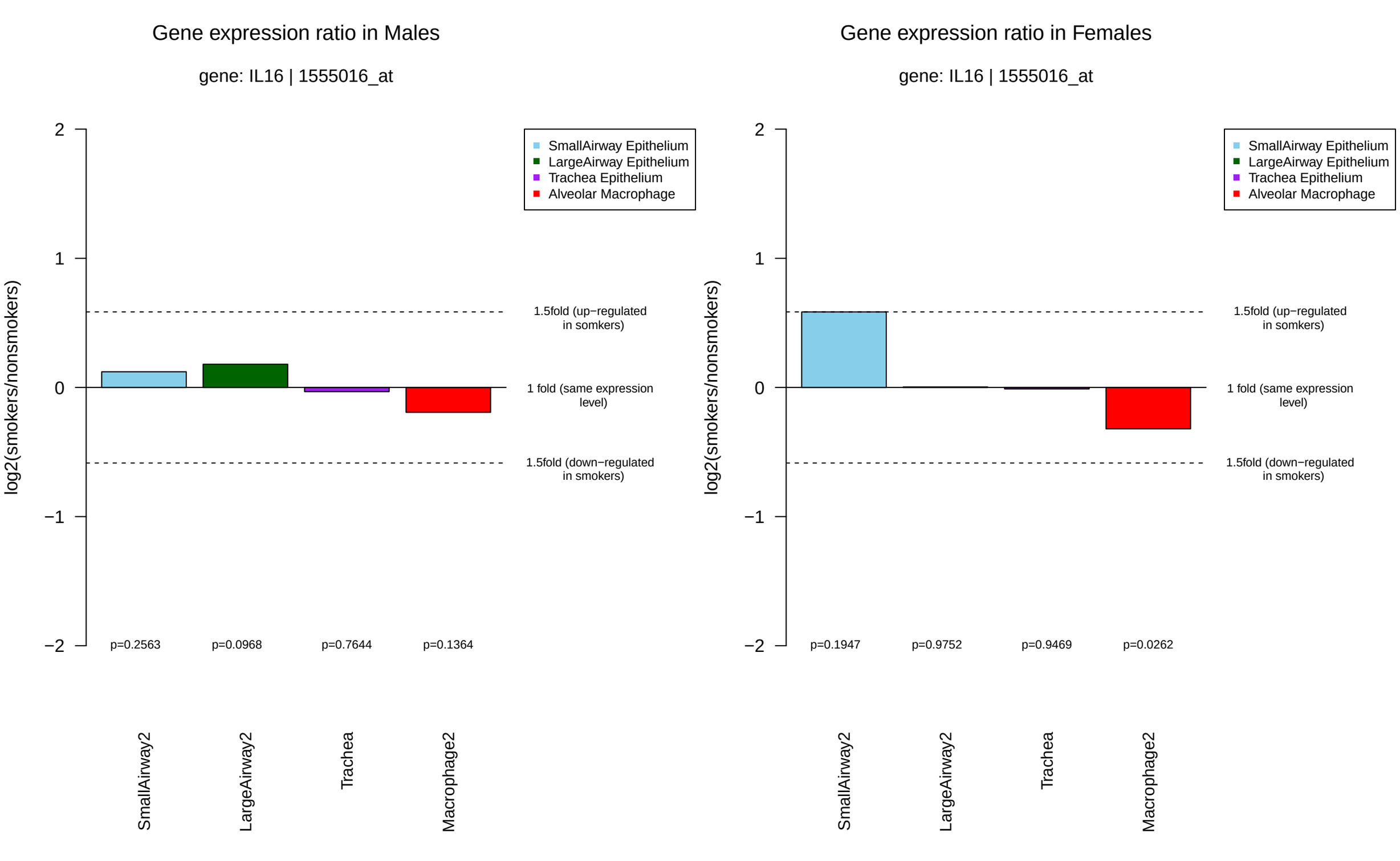
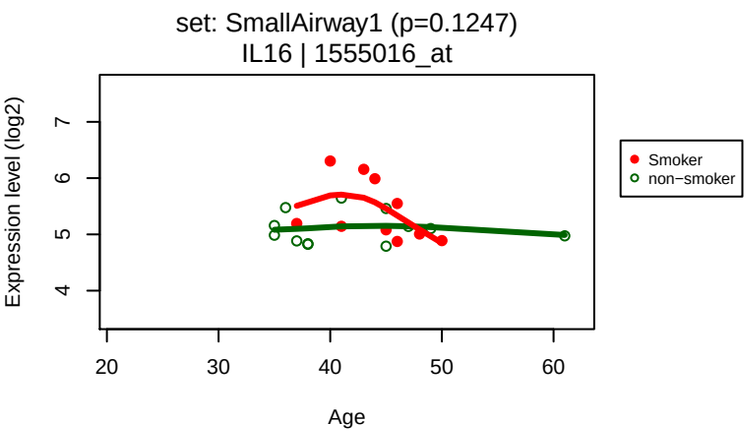
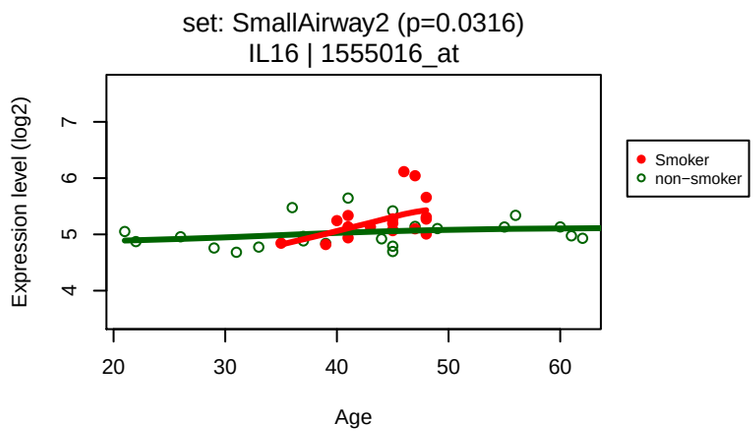
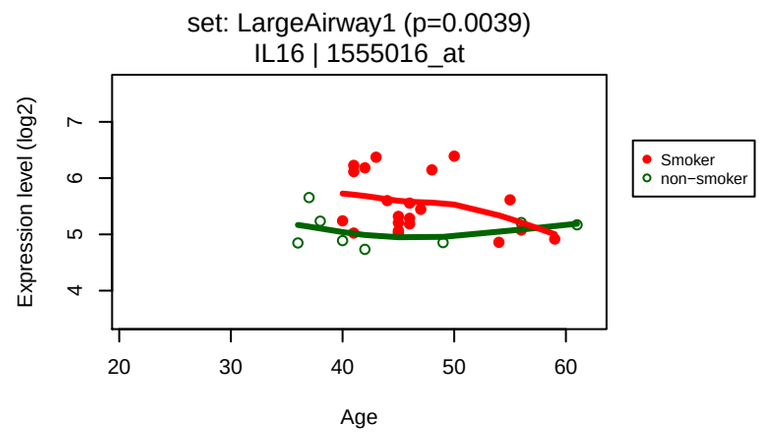

Gene expression of smokers and nonsmokers
gene: IL16 | 1555016_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1247
p=0.0316
p=0.0571
p=0.0039
p=0.1193
p=0.4619
p=0.5495
p=0.0173
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

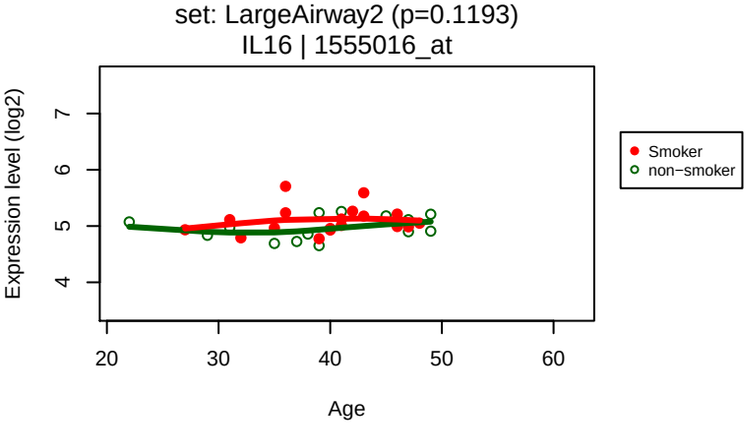

Gene expression ratio (smokers/nonsmokers)
gene: IL16 | 1555016_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1247
p=0.0316
p=0.0571
p=0.0039
p=0.1193
p=0.4619
p=0.5495
p=0.0173
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

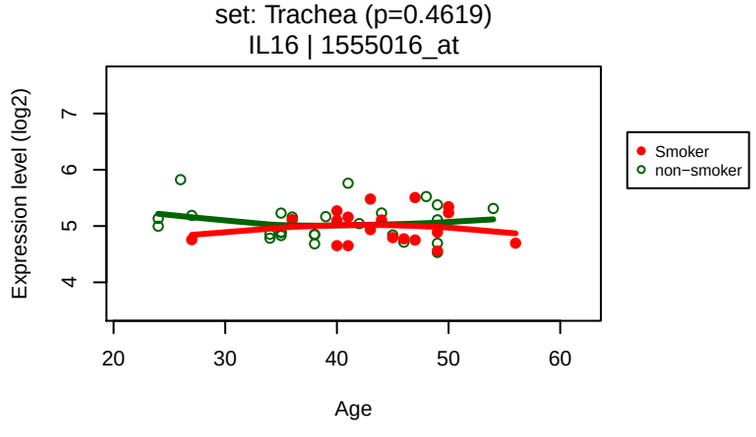

Gene expression ratio in Males
gene: IL16 | 1555016_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2563
p=0.0968
p=0.7644
p=0.1364
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IL16 | 1555016_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1947
p=0.9752
p=0.9469
p=0.0262
SmallAirway2
LargeAirway2
Trachea
Macrophage2

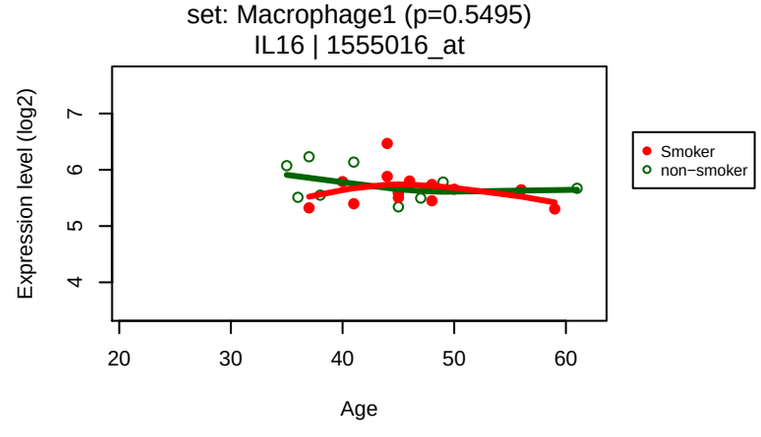

set: SmallAirway1 (p=0.1247)
IL16 | 1555016_at
Expression level (log2)
Age
Smoker
non−smoker

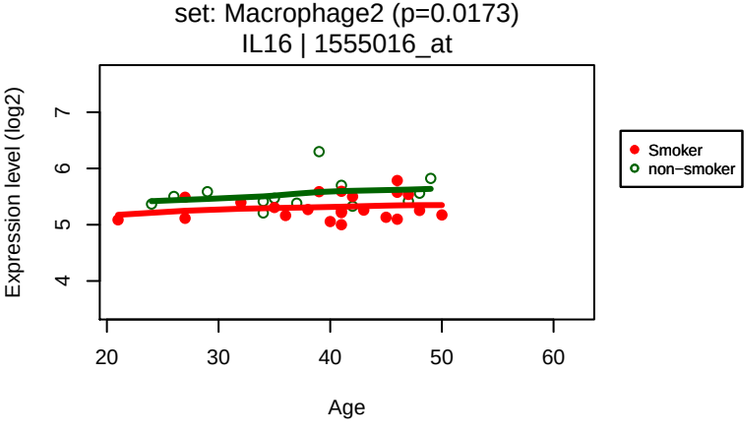

set: SmallAirway2 (p=0.0316)
IL16 | 1555016_at
Expression level (log2)
Age
Smoker
non−smoker

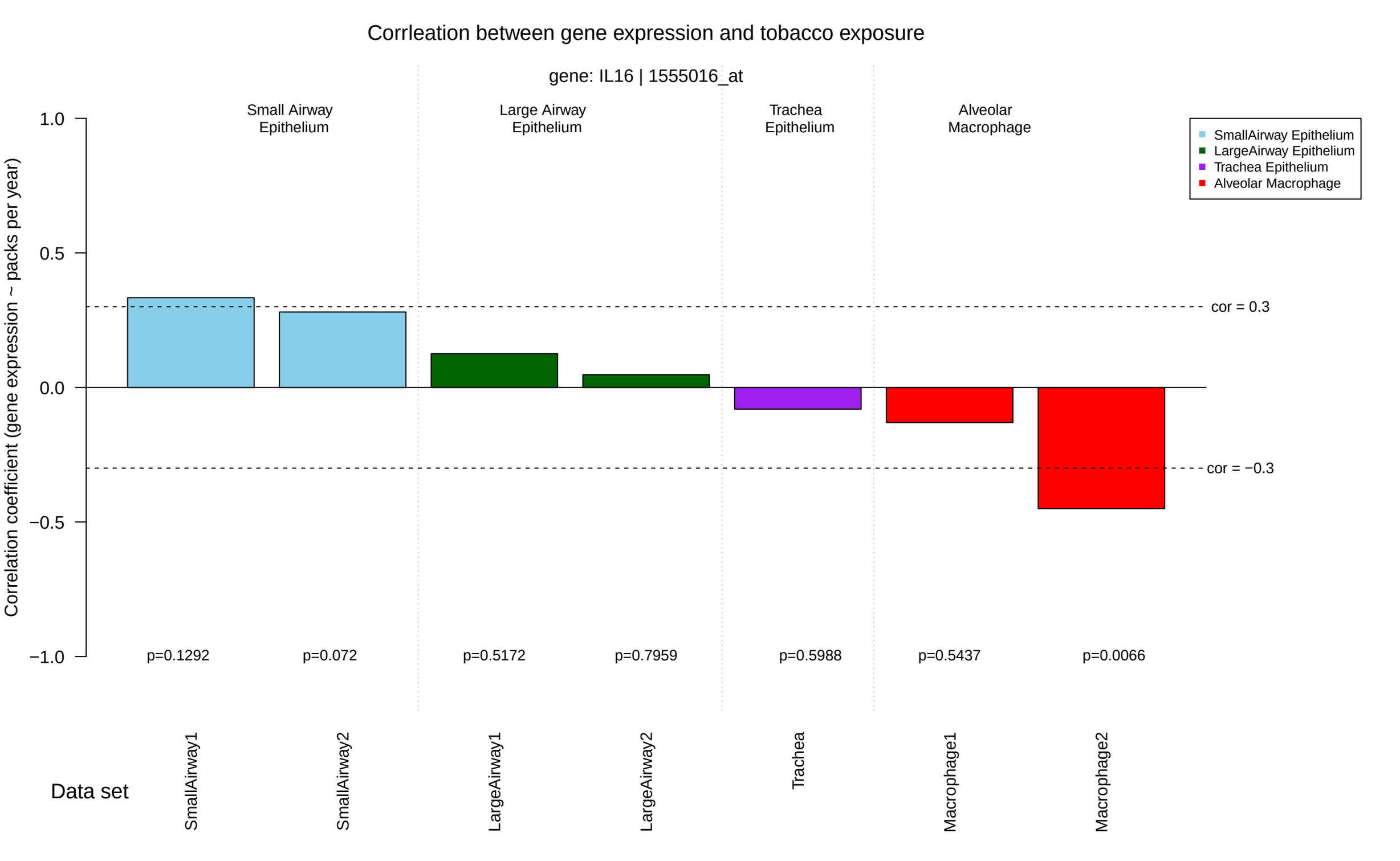

set: LargeAirway1 (p=0.0039)
IL16 | 1555016_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1193)
IL16 | 1555016_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4619)
IL16 | 1555016_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5495)
IL16 | 1555016_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0173)
IL16 | 1555016_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IL16 | 1555016_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1292
p=0.072
p=0.5172
p=0.7959
p=0.5988
p=0.5437
p=0.0066
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2